

Some proof writing guidance (adapted from Dana Ernst):

- You should indicate where the proof begins by writing “Proof.” at the beginning.
- Make it clear to yourself and the reader what your assumptions are at the very beginning of your proof. Typically, these statements will start off “Assume...”, “Suppose ...”, or “Let ...”. Sometimes there will be some implicit assumptions that we can omit, but at least in the beginning, you should get in the habit of clearly stating your assumptions up front.
- Carefully consider the order in which you write your proof. Each sentence should follow from an earlier sentence in your proof or possibly a result you have already proved.
- Unlike the experience many of you had writing proofs in your high school geometry class, our proofs should be written in complete sentences. You should break sections of a proof into paragraphs and use proper grammar.
- There will be many situations where you will want to refer to an earlier definition, problem, theorem, or corollary. In this case, you should reference the statement by number, but it is also helpful to the reader to summarize the statement you are citing.
- There will be times when we will need to do some basic algebraic manipulations. You should feel free to do this whenever the need arises. But you should show sufficient work along the way. In addition, you should organize your calculations so that each step follows from the previous. The order in which we write things matters. You do not need to write down justifications for basic algebraic manipulations (e.g., adding 1 to both sides of an equation, adding and subtracting the same amount on the same side of an equation, adding like terms, factoring, basic simplification, etc.).
- On the other hand, you do need to make explicit justification of the logical steps in a proof. As stated above, you should cite a previous definition, theorem, etc. when necessary.
- Similar to making it clear where your proof begins, you should indicate where it ends. It is common to conclude a proof with the standard “proof box” ($\square$ or $\blacksquare$). This little square at end of a proof is sometimes called a tombstone or Halmos symbol after Hungarian-born American mathematician Paul Halmos (1916–2006).

Theorem (Well Ordering Principle):

Important Definitions:

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Theorem (Division Algorithm). *Let a and b be unique integers, with $b \geq 0$, then*

How would you prove this? What are the important words?

Proof:

Theorem (The GCD is a linear combination). *For any nonzero integers a and b , there exist*

Exercise. Show that $5n + 3$ and $7n + 4$ are relatively prime for all n .

Exercise. Show that $\gcd(a, bc) = 1$ if and only if $\gcd(a, b) = 1$ and $\gcd(a, c) = 1$.

Exercise. Prove that every prime greater than 3 can be written in the form $6n + 1$ or $6n + 5$.

Lemma 1 (Euclid's Lemma). *If p is a prime that divides ab , then*

Proof:

Theorem (Fundamental Theorem of Arithmetic).

Recall:

- When $a = nq + r$ we write
- In general, if a and b are integers and n is a positive integer, then $a \pmod{n} = b \pmod{n}$ iff

Example. Determine $7^{1000} \pmod{6}$.

Example. Determine $6^{1001} \pmod{7}$.

Example. $8 \equiv 13 \pmod{?}$

Definition. *An equivalence relation on a set S is a set R of ordered pairs of elements of S such that:*

Example. Let S be the set of integers. If $a, b \in S$, define $a \sim b$ iff $a \pmod{n} \equiv b \pmod{n}$. Prove that $\sim$ is an equivalence relation and determine the equivalence classes of S

How many equivalence classes are there?